Brisk Tuning Guide (for Brisk 1.4)

This guide explains how to choose Brisk settings for common workloads. It applies to Brisk 1.4 and later. All recommendations were measured on an 8-core laptop.

Choosing a level. For log files and other data that is written once and rarely read, use level 3: it compresses at over 400 MB/s while saving most of the space that higher levels save. For database backups, use level 9, which is the best balance when backups are read back regularly. Use level 13 only for cold archives that are almost never read; it can use up to 1.5 GiB of memory during compression.

Dictionaries. Small files compress poorly because there is little repeated data inside each file. For collections of small files under 16 KiB, train a dictionary first with brisk train --samples <directory> --size 112K. The recommended dictionary size is 112 KiB; larger dictionaries rarely help and slow down decompression.

Threads. Brisk uses one thread per CPU core by default. Above 8 threads the speed gain is small, so on large servers set --threads 8 to leave cores free for other work.

Checking results. Run brisk bench --level <n> on a sample of your data before changing settings in production. The benchmark prints the compression ratio and the speed for each level.